

ORIGINAL ARTICLE

Endocrine-disrupting properties in vivo of widely used azole fungicides

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Summary

The endocrine-disrupting potential of four commonly used azole fungicides, propiconazole, tebuconazole, epoxiconazole and ketoconazole, were tested in two short-term in vivo studies. Initially, the antiandrogenic effects of propiconazole and tebuconazole (50, 100 and 150 mg/kg body weight/day each) were examined in the Hershberger assay. In the second study, pregnant Wistar rats were dosed with propiconazole, tebuconazole, epoxiconazole or ketoconazole (50 mg/kg/day each) from gestational day (GD) 7 to GD 21. Caesarian sections were performed on dams at GD 21. Tebuconazole and propiconazole demonstrated no antiandrogenic effects at doses between 50 and 150 mg/kg body weight/day in the Hershberger assay. In the in utero exposure toxicity study, ketoconazole, a pharmaceutical to treat human fungal infections, decreased anogenital distance and reduced testicular testosterone levels, demonstrating a demasculinizing effect on male fetuses. Tebuconazole, epoxiconazole and ketoconazole induced a high-frequency of post-implantation loss, and both ketoconazole and epoxiconazole caused a marked increase in late and very late resorptions. Overall the results show that many of the commonly used azole fungicides act as endocrine disruptors in vivo, although the profile of action in vivo varies. As ketoconazole is known to implicate numerous endocrine-disrupting effects in humans, the concern for the effects of the other tested azole fungicides in humans is growing.

Introduction

Epidemiological research has indicated a causal connection between human exposure to pesticides and endocrine-disrupting effects such as poor sperm quality (Swan *et al.*, 2003) and increased incidence of cryptorchidism (Weidner *et al.*, 1998).

Previously we found that many commonly used pesticides had endocrine-disrupting properties in one or more in vitro tests covering endocrine end points (aromatase activity, androgen, oestrogen or aryl hydrocarbon receptor binding). One of these pesticides was the imidazole fungicide prochloraz, which induced significant responses in all test systems (Andersen *et al.*, 2002; Long *et al.*, 2003). Subsequently, prochloraz was also found to induce antiandrogenic effects in rats in vivo in a Hershberger test as well as in developmental toxicity studies (Vinggaard *et al.*, 2002, 2005a; Blystone *et al.*, 2007). The purpose of the present study was to investigate the endocrine-

disrupting potential of four commonly used azole fungicides: propiconazole, tebuconazole, epoxiconazole and ketoconazole, and to compare the results obtained with those of prochloraz (Vinggaard *et al.*, 2005a; Laier *et al.*, 2006). Recently, tebuconazole was found to be endocrine-disrupting and epoxiconazole to be fetotoxic (Taxvig *et al.*, 2007) and more details on these findings were needed. The tested fungicides belong to a large group of compounds called conazoles, which includes the imidazole and triazole fungicides. Propiconazole, tebuconazole and epoxiconazole are triazoles, used in large amounts in the control of fungi in grain and to a lesser extent in vegetable and fruit production. Ketoconazole is an orally active imidazole antifungal drug used for treatment of systemic infections and is known to inhibit testicular androgen production both in vitro and in vivo (Sikka *et al.*, 1985; De Coster *et al.*, 1986). Ketoconazole was included in the study, because of its known hormone-disrupting effects in humans and thereby a correlation

between human effects and animal data could be drawn. Initially, the antiandrogenic effects of propiconazole and tebuconazole were examined in the *in vivo* Hershberger assay. Secondly, developmental effects of the four fungicides were investigated by dosing pregnant rats from gestational day (GD) 7 to GD 21, followed by examination of the dams and the fetuses. This later study was performed to investigate endocrine-disrupting effects in dams, but in particular in fetuses, which are considered very sensitive. In this study developmental effects in both female and male fetuses were studied, whereas the Hershberger assay, in this case, was designed to look for antiandrogenic effects in young adult male rats only.

Materials and methods

Test compounds

The following test compounds were used: propiconazole 97.5% (CAS no. 60207-90-1; Dr Ehrendorfer EHRC 16480000), tebuconazole 98% pure (CAS no. 107534-96-3; Dr Ehrendorfer EHRC 17178700), epoxiconazole 99% pure (CAS no. 106325-08-0; Dr Ehrendorfer EHRC 13185000) and ketoconazole 99% pure (CAS no. 65277-42-1; Dr Ehrendorfer EHRC 14532000) from Bie & Berntsen (Rødovre, Denmark). The vehicle in the *in utero* exposure study was corn oil from Sigma-Aldrich (Brønby, Denmark).

In the Hershberger study, testosterone propionate (CAS no. 57-85-2; Nomeco A/S, Herlev, Denmark) and flutamide (CAS no. 13311-84-7; Sigma-Aldrich, St Louis, MO, USA) were employed. All test compounds were dissolved in peanut oil (no. P-2144; Sigma-Aldrich). Sterile peanut oil (The Royal Veterinary Agriculture Pharmacy, Copenhagen, Denmark) was used for testosterone propionate and flutamide solutions. All compounds were administered in a dosing volume of 2 mL/kg body weight.

Hershberger test

For assessment of antiandrogenicity, the fungicides were administered to castrated males treated with testosterone as previously described (Birkhoj *et al.*, 2004; Vinggaard *et al.*, 2005b) and the effects were examined in both intact and castrated testosterone-treated males by analysing weights of reproductive organs, serum hormone levels and prostate gene expression as described (Vinggaard *et al.*, 2005b).

One group of intact animals and nine groups of castrated male Wistar rats were included in the study ($n = 6$ per group). The intact rats and the one group of castrated rats served as negative controls and were given peanut oil only. The rats in the remaining eight groups were all dosed with testosterone propionate (0.5 mg/kg/day *s.c.*).

One group was only administered testosterone propionate and served as the control group to which all other groups were compared. The positive control group received flutamide (10 mg/kg body weight/day orally). The last six groups received propiconazole or tebuconazole orally at doses of 50, 100 or 150 mg/kg body weight/day.

Real-time RT-PCR

Ventral prostate cDNA samples were quantified on the 7900HT Fast Real-Time PCR System (Applied Biosystems, Foster City, CA, USA) by standard TaqMan technology and using the primer and probe sets as described in Laier *et al.* (2006). Expression levels of the following genes were quantified: the prostate binding protein C3 (PBP C3), ornithine decarboxylase (ODC), testosterone-repressed prostate message 2 (TRPM-2), complement component 3 (Compl.C3), the androgen receptor and insulin-like growth factor 1. These genes were chosen because they have previously been shown to be androgen-regulated or to be valuable for investigating antiandrogenic effects in the Hershberger assay (Nellemann *et al.*, 2001, 2005).

Hormone levels

In the Hershberger assay, rat lutenizing hormone, rat follicle-stimulating hormone and T4 hormone levels were analysed as previously described in Vinggaard *et al.* (2002).

In utero exposure toxicity study

Animals and exposure

A total of 50 time-mated, young adult Wistar rats (HanTac; WH, Taconic M&B, Ejby, Denmark) were supplied at day 3 of pregnancy (the day after mating defined as GD 1). The animals were distributed, housed and handled as previously described in Vinggaard *et al.* (2005a). The day after arrival, i.e. GD 4, animals were weighed and assigned to five groups of 10 animals per group, with similar weight distributions. An acclimatization period of 4 days was allowed before starting exposure. The rats were gavaged with vehicle (corn oil) or 50 mg/kg epoxiconazole, 50 mg/kg ketoconazole, 50 mg/kg propiconazole or 50 mg/kg tebuconazole, respectively, from GD 7 to GD 21. The dams were given a dosing volume of 2 mL/kg body weight.

Health status of dams

In the five groups of 10 animals, six, eight, nine, eight and nine rats were pregnant respectively. The females were observed daily for signs of toxicity. Body weights were recorded on GD 4 and daily during the entire dosing period. The maternal weight gain from GD 7 to GD 21 was calculated.

Caesarian sections GD 21

At GD 21 dams were weighed, anaesthetized in CO₂/O₂ and decapitated and caesarian sections were performed as previously described in Vinggaard *et al.* (2005a). Trunk blood was collected into heparin-coated vials from all fetuses for hormone analysis; one pool per litter was made for all male and female fetuses. Body weight and anogenital distance (AGD) was measured for all fetuses.

Hormone analysis

In the in utero exposure toxicity study, testosterone and progesterone were analysed in serum from dams and fetuses at GD 21, as were ex vivo testosterone and progesterone productions as described in Laier *et al.* (2006). Serum 17 α -hydroxyprogesterone levels were analysed by use of a 17 α -hydroxyprogesterone-enzyme immunoassay kit from Assay Designs according to the manufacturer's instructions (Electra-Box Diagnostica Aps, Rødovre Denmark). Steroid hormones were analysed in testis and oestradiol in ovaries as described in Taxvig *et al.* (2007).

Statistical analyses

For statistical evaluation of plasma hormone levels a one-way analyses of variance (ANOVA) was employed for all groups and, if significant, followed by the *post hoc* Dunnett's test. Significance was judged at $p < 0.05$. For statistical evaluation of pregnancy and litter data, the litter was considered as the statistical unit and the alpha level was 0.05. The results were analysed using ANOVA and to adjust for litter effects, litter was included in the analysis of variance as a nested factor.

Results**The Hershberger test**

The azole fungicides propiconazole and tebuconazole (50, 100 and 150 mg/kg body weight/day each) were investigated for antiandrogenic effects in castrated testosterone-treated male rats. The body and organ weights were unaffected by the treatments except for an increase in liver weights caused by propiconazole at 100 and 150 mg/kg body weight/day and by 150 mg/kg body weight/day tebuconazole (Table 1). The positive control flutamide reduced weights of ventral prostate, seminal vesicles/coagulation gland, levator ani/bulbocavernosus muscles (LABC) and bulbourethral glands as expected for an anti-androgen (Table 1).

Regarding hormone levels, the only effect observed was an increase in serum FSH after treatment with 150 mg/kg body weight/day propiconazole, whereas the positive control flutamide increased both serum LH and FSH levels as expected, while T4 was unaffected (Table 1).

Table 1. Body weight, organ weights and serum hormone levels

	Intact control	Castrated control	Castrated rats given testosterone propionate								
			Control	Flutamide	Tebu 50	Tebu 100	Tebu 150	Propi 50	Propi 100	Propi 150	
Body weight (g)	183 ± 14	166 ± 3	165 ± 10	165 ± 18	164 ± 16	165 ± 14	170 ± 15	170 ± 15	171 ± 21	173 ± 10	
Prostate (mg)	87.5 ± 10.8*	5.7 ± 3.6*	49.2 ± 18.0	11.1 ± 5.9 ^a	57.2 ± 13.6	51.4 ± 10.8	61.0 ± 8.8 ^a	53.8 ± 15.2	43.0 ± 13.7	59.2 ± 12.9	
Seminal vesic. (mg)	114.0 ± 36.7	12.5 ± 3.0*	95.9 ± 39.7	20.1 ± 3.9*	106.2 ± 12.4	90.5 ± 7.7	101.4 ± 11.1	92.4 ± 13.3	96.5 ± 20.1	88.5 ± 22.4	
Musc. lev. ani (mg)	229 ± 24	93 ± 11*	199 ± 12	124 ± 24*	202 ± 38	216.7 ± 22.6	222 ± 27	208.9 ± 25	204 ± 38	221 ± 23	
Bulbourethral gl. (mg)	15.8 ± 4.1*	1.7 ± 0.7 ^a	9.0 ± 2.2	1.8 ± 0.8*	10.5 ± 1.3	10.4 ± 1.3	10.7 ± 2.0	10.2 ± 1.7	9.2 ± 1.9 ^a	10.9 ± 1.5	
Pituitary (mg)	6.2 ± 0.5	7.6 ± 1.0	6.4 ± 0.6	6.6 ± 1.9	6.5 ± 0.7	6.7 ± 0.9	7.5 ± 1.0	6.9 ± 0.6	7.4 ± 0.7	7.5 ± 0.7	
Thyroid (mg)	9.5 ± 3.8	9.1 ± 5.0	8.0 ± 2.0	8.3 ± 1.1	8.0 ± 1.0	8.2 ± 1.8	8.4 ± 3.0	8.1 ± 1.8	7.7 ± 2.7	7.3 ± 1.4	
Liver (g)	8.2 ± 0.6*	7.2 ± 0.3	6.9 ± 0.5	7.7 ± 0.9	7.2 ± 0.8	7.4 ± 0.7	8.5 ± 1.0*	7.9 ± 0.9	8.2 ± 1.1*	8.5 ± 0.5*	
Kidney (g)	1.5 ± 0.1 ^a	1.3 ± 0.1	1.3 ± 0.1	1.3 ± 0.1	1.2 ± 0.1	1.3 ± 0.1	1.4 ± 0.2	1.4 ± 0.1	1.4 ± 0.2	1.4 ± 0.1	
Serum hormone levels											
LH (ng/mL)	1.20 ± 0.54	2.0 ± 18.9*	0.8 ± 1.3	15.8 ± 6.2*	0.9 ± 0.8	0.5 ± 0.6	1.0 ± 1.0	1.6 ± 1.2	1.5 ± 2.8	1.9 ± 1.9	
FSH (ng/mL)	9.80 ± 0.97*	32.2 ± 5.6*	20.8 ± 3.2	36.9 ± 5.0*	24.2 ± 3.7	19.7 ± 5.8	21.6 ± 3.5	21.5 ± 7.8	24.1 ± 6.9	32.7 ± 0.8*	
T4 (nm)	127 ± 41	110 ± 28	140 ± 41	123 ± 23	145 ± 33	141 ± 39	146 ± 42	122 ± 29	116 ± 29	127 ± 29	

Data represents mean $\pm$ SEM ($n = 6$). * $n = 5$; one-way anova. *Statistical significance compared to controls by Dunnett's test ($p < 0.05$).

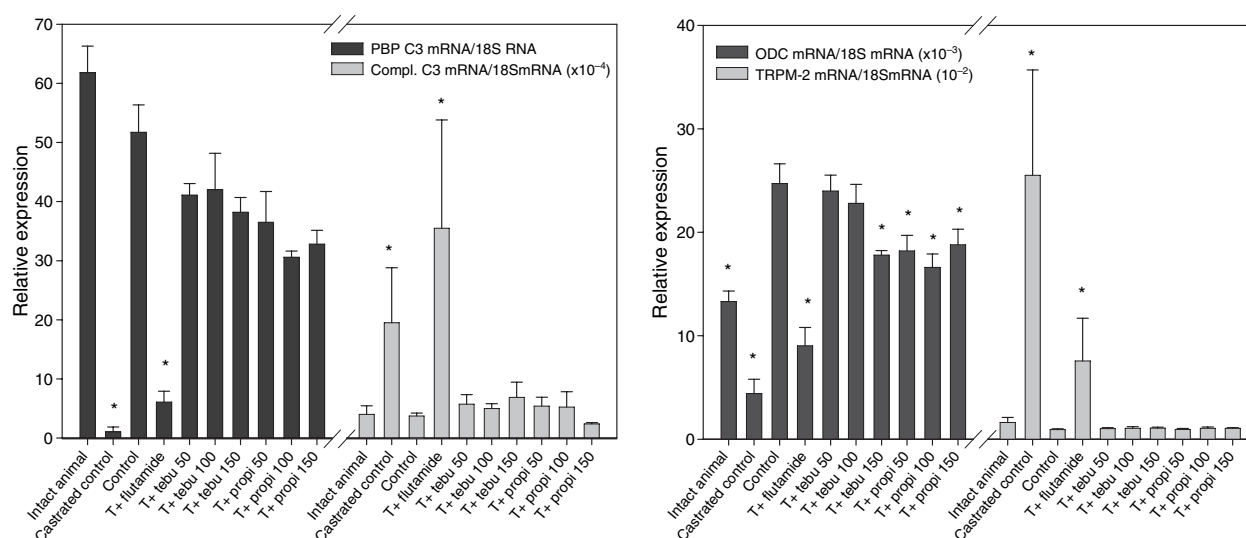


Figure 1 PBP C3, ODC, TRPM-2 and Compl.C3 mRNA expressed relative to the housekeeping gene 18S rRNA in ventral prostate from castrated rats treated for 7 days with testosterone propionate (T) (0.5 mg/kg/ s.c.) with or without flutamide (10 mg/kg orally) or with tebuconazole (tebu) or propiconazole (propi) (50, 100 and 150 mg/kg orally). The rats in the group labelled 'Control' were given testosterone propionate alone, and this group served as the control group to which all groups were compared. The 'Intact animal' as well as the 'Castrated control' was given oil only. Data represent the mean $\pm$ SEM ($n = 6$). *Indicates statistically significant difference compared to controls (castrated testosterone-treated animals) ($p < 0.05$).

The gene expression levels in the ventral prostate of PBP C3, TRPM-2 and Compl.C3 were unaffected by azole fungicide treatment, whereas ODC was significantly decreased by all doses of propiconazole and by the highest dose of tebuconazole. The antiandrogen flutamide, which was given as a positive control, had a statistically significant effect on the expression of all genes, as expected and caused qualitatively the same effects as seen after castration (Fig. 1).

In utero toxicity testing

The effects of propiconazole, tebuconazole, epoxiconazole and ketoconazole were investigated by dosing pregnant dams during gestation followed by examination of fetuses for effects on sexual differentiation.

Pregnancy and litter data

Ketoconazole induced a significant reduction in body weight of the dams, as well as a decrease in the number of live fetuses (Table 2). Tebuconazole, epoxiconazole and ketoconazole induced a high frequency of post-implantation loss and epoxiconazole also caused a marked increase in late and very late resorptions. Propiconazole and epoxiconazole caused increased fetal weight in both sexes, although for propiconazole the difference was only significant in male fetuses (Table 2).

Anogenital distance

Ketoconazole significantly decreased the AGD in both male and female fetuses (Table 2). No significant effects on AGD of propiconazole, tebuconazole or epoxiconazole were observed.

Hormone levels

In dams, a decrease in the plasma oestradiol level was seen in the tebuconazole-, epoxiconazole- and ketoconazole-exposed groups. Epoxiconazole led to an increase in the testosterone level (Table 3). Furthermore, in propiconazole-exposed dams a significant increase in 17α -hydroxyprogesterone was observed (Table 3).

Testicular testosterone and progesterone were analysed in male fetuses. Tebuconazole caused a significant increase in progesterone levels and testosterone levels were significantly decreased by ketoconazole (Table 3). Propiconazole and epoxiconazole had no significant effect on the measured hormone levels in fetuses (Table 3). No significant effects were found on ex vivo testicular testosterone and progesterone production; however, a tendency towards a decrease in testosterone production was seen in all four groups, especially in the ketoconazole group (Table 3). No changes were observed in fetal plasma testosterone and progesterone levels or in the oestradiol levels in ovaries (data not shown).

Table 2 Pregnancy and litter data

GD21 Caesarian section	1. Control	2. Propi-50	3. Tebu-50	4. Epoxi-50	5. Keto-50
No. dams	6	8	9	8	9
Maternal body weight	224.0 ± 8.3	218.5 ± 5.2	224.7 ± 4.5	218.2 ± 3.9	200.8 ± 5.8 **
No. implantations	11.3 ± 0.8	12.3 ± 0.4	11.3 ± 0.9	11.9 ± 0.5	9.2 ± 1.6
No. live fetuses	11.3 ± 0.8	12.0 ± 0.4	10.4 ± 1.2	9.0 ± 1.1	4.9 ± 1.4 **
% post-implantation loss	0.0 ± 0.0	2.0 ± 1.3	11.4 ± 6.2 *	25.1 ± 8.0 *	47.7 ± 12.7 **
% late res	0.0 ± 0.0	0.0 ± 0.0	4.4 ± 4.4	22.4 ± 7.9 *	45.8 ± 12.8 **
% very late res	0.0 ± 0.0	0.0 ± 0.0	2.2 ± 2.2	15.5 ± 5.6 *	22.5 ± 7.6 *
% males	56.5 ± 2.6	52.3 ± 2.0	48.3 ± 7.6	61.1 ± 6.3	63.0 ± 10.5
Male fetal weight ^a	3.6 ± 0.2	4.5 ± 0.3*	3.9 ± 0.3	4.8 ± 0.3*	3.9 ± 0.3
Female fetal weight ^a	3.5 ± 0.2	4.2 ± 0.3	3.6 ± 0.2	4.3 ± 0.3*	3.5 ± 0.3
Male AGD (mm) ^b	3.76 ± 0.08	3.87 ± 0.06	3.75 ± 0.05	3.97 ± 0.06	3.46 ± 0.15**
Male AGD index	2.47 ± 0.03	2.37 ± 0.06	2.41 ± 0.07	2.39 ± 0.04	2.20 ± 0.07**
Female AGD (mm) ^b	2.12 ± 0.03	2.15 ± 0.06	2.12 ± 0.05	2.07 ± 0.04	1.90 ± 0.05**
Female AGD index	1.40 ± 0.03	1.35 ± 0.05	1.40 ± 0.05	1.28 ± 0.03	1.26 ± 0.03*

Propi-50, propiconazole 50 mg/kg/day; Tebu-50, tebuconazole 50 mg/kg/day; Epoxi-50, epoxiconazole 50 mg/kg/day; Keto-50, ketoconazole 50 mg/kg/day; res, resorption i.e. regression of the fetus; AGD, anogenital distance. Data represent group means, based on litter mean ± SEM.

^aFetal body weight was analysed using the live number of fetuses as a covariate. ^bAGD was analysed both with and without the cubic root of body weight as a covariate and both analyses showed that only group 5 was significantly affected. AGD index means the AGD divided by the cubic root of body weight. Values shown in bold are statistically significantly different compared with control, **p* < 0.05 and ***p* < 0.01 respectively.

Table 3 Plasma hormone levels in dams at GD 21 and testicular hormone levels in fetuses

	17 α -Hydroxyprogesterone (ng/mL)	Progesterone mean (nm)	Testosterone mean (nm)	Oestradiol mean (nm)
Control	13.11 ± 4.02 (8)	95 ± 100 (9)	0.38 ± 0.32 (9)	0.036 ± 0.024(9)
Propiconazole 50 mg/kg	33.37 ± 12.95*(7)	65 ± 71 (9)	0.57 ± 0.26 (9)	0.029 ± 0.011(10)
Tebuconazole 50 mg/kg	17.85 ± 9.87 (8)	161 ± 126 (10)	0.29 ± 0.19 (10)	0.015 ± 0.006*(10)
Epoxyconazole 50 mg/kg	23.77 ± 10.41 (9)	175 ± 102 (10)	0.99 ± 0.57*(10)	0.021 ± 0.006*(10)
Ketoconazole 50 mg/kg	6.75 ± 3.56 (10)	32 ± 29 (9)	0.13 ± 0.12 (9)	0.015 ± 0.004*(10)
	Progesterone (ng/testis)	Testosterone (ng/testis)	Testosterone production (ng/testis)	Progesterone production (ng/testis)
Control	0.08 ± 0.04 (16)	1.64 ± 0.68 (13)	4.90 ± 3.80 (6)	0.001 ± 0.00 (6)
Propiconazole 50 mg/kg	0.10 ± 0.06 (21)	1.79 ± 0.81 (19)	3.10 ± 2.70 (8)	0.003 ± 0.001 (8)
Tebuconazole 50 mg/kg	0.27 ± 0.12*(23)	1.38 ± 0.88 (23)	2.50 ± 1.60 (8)	0.004 ± 0.002 (8)
Epoxyconazole 50 mg/kg	0.06 ± 0.04 (16)	1.42 ± 0.81 (17)	2.40 ± 1.10 (8)	0.003 ± 0.003 (8)
Ketoconazole 50 mg/kg	0.07 ± 0.00 (14)	0.35 ± 0.44*(14)	1.10 ± 0.60 (6)	0.004 ± 0.002 (6)

Testes from male fetuses (GD 21) exposed to propiconazole, tebuconazole, epoxiconazole or ketoconazole (50 mg/kg) were extracted with diethyl ether and hormone levels were analysed as described in Materials and methods. Data represent mean ± SD. Values shown in bold are statistically significantly different compared with control, **p* < 0.05. () = *n*.

Discussion

The overall objective of this study was to identify potential common modes of actions for the endocrine disruption effects of azole fungicides.

The imidazole prochloraz has previously been demonstrated to be antiandrogenic in adult male rats and to disturb reproductive development in the offspring after perinatal exposure (Noriega *et al.*, 2005; Vinggaard *et al.*, 2005a; Laier *et al.*, 2006). Tebuconazole and propiconazole

have demonstrated antiandrogenic activity in vitro (Birkhøj, M., Taxvig, C., Vinggaard, A.M., Nellesmann, C. and Andersen, H.R., unpublished data), but in contrast to prochloraz, that induced marked antiandrogenic effects at doses between 50 and 150 mg/kg body weight/day in the Hershberger assay, no antiandrogenic effects were found for tebuconazole and propiconazole in this test. At 150 mg/kg body weight/day, propiconazole increased FSH levels and both compounds caused a down-regulation of ODC mRNA in ventral prostates at this dose level.

ODC is an androgen-regulated gene but is regulated by other pathways as well, including regulation by oestrogens (Nellemann *et al.*, 2005). As none of the other end points, such as weight of androgen-dependent organs, expression of androgen-responsive genes or serum levels of LH were affected, we believe that the observed effects are the result of a non-androgen-regulated pathway. Liver weights were increased with the highest doses of both tebuconazole and propiconazole, as observed previously for prochloraz. A stimulation of liver growth is often observed with chemicals that induce microsomal liver enzymes like cytochrome P450 enzymes (Juberg *et al.*, 2006).

The results from this study demonstrate a demasculinizing effect of ketoconazole on male fetuses, as both decreased AGD and reduced testicular testosterone levels were observed. Furthermore, ketoconazole also resulted in a decreased AGD in female fetuses. The mechanism behind this effect of ketoconazole on female fetuses is unknown, but looking at the plasma hormone levels measured in the dams, a tendency towards a decrease in all measured hormone levels is observed. An overall disturbance of the steroid hormone balance at the level of cholesterol could be the cause of these disturbances. This would agree with studies showing that ketoconazole inhibits cholesterol synthesis (Kan *et al.*, 1985). No significant effect on AGD was observed for any of the other tested fungicides in the current study. The observed effects of ketoconazole on male fetuses are in good agreement with previous *in vivo* studies where ketoconazole has been found to inhibit testicular function and male reproductive function, e.g. by causing spermatid retention and decreased serum testosterone levels in male rats (Adams *et al.*, 1998; O'Connor *et al.*, 2002). In humans, ketoconazole has been found to cause decreases in plasma testosterone and cortisol levels (Pont *et al.*, 1982a,b), resulting in gynaecomastia and oligospermia in men and menstrual irregularities in women (Thompson & Carter, 1993).

Tebuconazole caused a significant increase in testicular progesterone levels, in agreement with our recent study, where tebuconazole also caused a significant increase in testicular progesterone and 17 α -hydroxyprogesterone levels and a decreased testicular testosterone level at GD 21, indicating demasculinization of male fetuses (Taxvig *et al.*, 2007). No demasculinizing effects on male fetuses were seen with epoxiconazole or propiconazole administration. In contrast, an increased fetal weight was observed, which might be related to the up-regulated levels of testosterone observed in the dams (only statistically significant for epoxiconazole-exposed dams). For epoxiconazole, the increased maternal testosterone level in plasma as well as the reduced oestradiol level, is in agreement with studies reported as part of the Danish approval of epoxiconazole, where increased serum concentrations of

testosterone, androstendion, FSH and ACTH as well as reduced oestradiol, corticosterone and aldosterone was reported in adult male rats after oral administration of approximately 250 mg/kg body weight/day (3000 p.p.m. in the feed) for 28 days. The marked increase in post-implantation loss as well as late resorptions is in agreement with results of a previous study, where epoxiconazole was found to be markedly fetotoxic (Taxvig *et al.*, 2007). Overall the results from our studies reveal that four of the commonly used azole fungicides act as endocrine disruptors *in vivo*. However, the profile of action *in vivo* in developmental toxicity studies varied. The imidazoles prochloraz and ketoconazole show more endocrine-disrupting activity *in vivo* and have a profile of action that is more alike than for the triazoles. Both imidazoles cause demasculinization of the male offspring or fetuses after perinatal exposure (Noriega *et al.*, 2005; Vinggaard *et al.*, 2005a; Laier *et al.*, 2006), but tebuconazole has also been found to demasculinize the male offspring and to possess some of the same endocrine effects as prochloraz (Taxvig *et al.*, 2007). Epoxiconazole is clearly fetotoxic, an effect now also observed for ketoconazole with increases in post-implantation loss and late/very late resorptions. Based on the many endocrine-disrupting effects of ketoconazole in man, there are reasons for concern regarding the human impacts of the partly similar azole fungicides used within gardening and agriculture.

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Panel discussion

P. Foster

Azole fungicides are used extensively in grape vines. They were developed to inhibit sterol biosynthesis in fungi and it is not surprising that they inhibit steroidogenesis in animals. They are potent P450 inhibitors in fungi and I assume that is their mode of action as antisteroidogenesis in animals.

C. Nellemann

Yes, they inhibit P450 and have other effects on steroid biosynthesis.

H. Leffers

Denmark is now an approved wine producing country. Many fungicides such as vinclozolin have antiandrogenic effects: is this related to their action on fungi? Fungi do not have androgen receptors (AR), but do the fungicides affect animal AR?

C. Nellemann

I do not know what is the action on AR. They have an effect on biosynthesis and inhibit P450 in the fungal mem-

brane. This is not a specific effect and will block other P450-dependent enzyme pathways. It is unclear why the preferred anti fungal agents only have antiandrogenic activity, and this may be coincidental. We assume that all fungicides are endocrine disrupters until proved not to be.

T. Søbørg

Ketoconazole is a drug in clinical use and therefore will have been exhaustively tested: its actions will be available from the producer.

C. Nellemann

We considered the known effects of ketoconazole in our studies, but results vary from laboratory to laboratory, therefore we repeated the tests in our own experiments.

N.E. Skakkebæk

We have not focused on adrenal effects at this Workshop but we know from medical endocrinology that ketoconazole affects the adrenal gland. Perhaps abnormalities of adrenocortical steroid biosynthesis should be included in our studies of hormone disruption.

C. Nellemann

We use a human adrenal cell line H295R to study steroid biosynthesis *in vitro*, and this shows many effects of different treatments. We would like to look more into the effects of endocrine disruption on the adrenals.

P. Foster

Have you seen reproductive tract malformations following treatment with these fungicides?

C. Nellemann

We have not seen hypospadias, cryptorchidism or differences in anogenital distance in fungicides other than prochloraz. They are less potent than prochloraz and this may be a dose effect. We use prochloraz as a positive control.

A.-M. Andersson

What is the mechanism of virilisation of female pups exposed *in utero* to antiandrogens?

C. Nellemann

This is also seen with other drugs but we do not know the mechanism of increased AGD observed with several of the fungicides. I would like to look at female reproductive tract development in more detail. We have found that other antiandrogens do not have this masculinising effect on female offspring.

A. Kortenkamp

Some of the Azole fungicides are AR antagonists in your *in vitro* studies but have no effect on genital tract development *in vivo*. Can you explain this apparent lack of correlation?

C. Nellemann

There is no direct comparison between *in vitro* and *in vivo* results. When there is evidence of an *in vitro* antiandrogen effect this does not give an indication of the *in vivo* potency. We were surprised when some of our antiandrogenic fungicides were negative in the Hersberger *in vivo* assay.